

Time Warp Tech — Journey Across Computer Eras

Game-based activity · evolution of computers

ID time-warp-tech

Subject Computer Science

Time 40 min

Type Game-based (5E)



Objectives

2 min

- Demonstrate an understanding of the evolution of computers across different generations. Understand
- Identify key characteristics and features of computers from various historical periods. Understand
- Apply critical thinking to analyse the technological advancements in each generation. Analyse



The activity & game

Mode: game-based. Students arrange computers from different generations onto the correct year-of-release boxes, matching each computer with its era of innovation.

- **Game:** “History of Computers” by Planeta 42
- **Open:** <http://planeta42.com/it/computerhistory/game.html>
- Click Start to play.
- Drag & drop each computer generation onto its year-of-release box.
- Complete the timeline in the minimum number of tries.



Introduction

8 min

Let's travel through time and see how computers grew from room-sized machines to the device in your pocket!



Engage — watch: Play a short video / slideshow of the evolution of computers: <https://youtu.be/asgjdkltB1c>

Engage — think: Pose a thought-provoking question: “Can you imagine a world without computers?” Let a few students share their ideas.



Explanation & Teaching

12 min

Explain

Present the history of computer generations — the pivotal innovations of each era and their impact on society.

Game mechanics

Introduce the game, underlining its dual purpose: it entertains and it teaches at the same time.

Elaborate

Students collaboratively explore the historical significance of each generation's breakthroughs, building a deeper appreciation for technological evolution.

Check for understanding

Which generation introduced personal desktop computers?

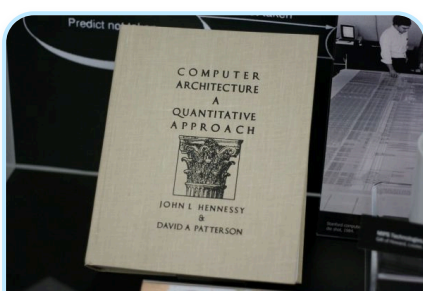
Later generations (microprocessor era) — students infer from the timeline.

Why did computers get smaller over time?

Smaller, faster components (vacuum tubes → transistors → chips).



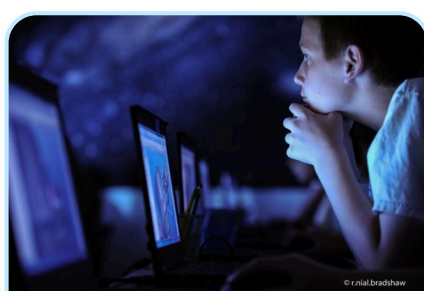
Generations of computers — real photos



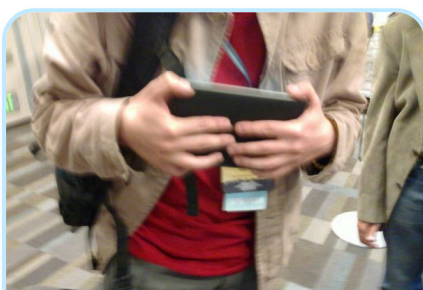
Early computers
room-sized machines



Desktop PC
personal computers



Laptop
portable



Tablet
touch, handheld



Explore — play the game

12 min

Students start playing: drag each computer to its release year. Before dropping, they discuss the historical aspect of that computer's era. Encourage students to complete the timeline in the fewest tries.

- Discuss the era before each drag & drop.
- Aim for the minimum number of tries.
- Work together — compare reasoning about each generation.



Evaluate & assessment

6 min

Thumbs up for True, thumbs down for False.

Early computers were much larger than today's laptops.



Every computer generation used the same technology.



New generations were driven by changing needs and better components.



Exit ticket

Students create a short historical record outlining the evolution of computer technology — how advancements were driven by changing needs and circumstances.

Homework

Formative assessment (Google Form): <https://forms.gle/NXD7qiVQ25TLuCY7>